- “classes” of HVDC power supplies depending on design constraints and where we want to start out
    - >40KV extremely high voltage power supply
        - allows the widest gap distance (50-100mm), dual polarity increases current efficiency, ~3mA current capacity, works well with multiple phases of coronal discharge
    - 40KV high voltage supply, the standard premium supply
        - allows up to 50mm gap distance, dual polarity increases current efficiency, ~3mA current capacity, works well with multiple phases of coronal discharge
    - 20KV high voltage supply, the most balanced option
        - allows up to 25mm gap distance, can be combined with a reversed polarity identical module to be upgraded to the 40KV dual polarity system once more finding is procured, ~1-3mA current capacity, a bit of a stretch for multi-stage
        - requires more consideration of insulated experimental environment
        - If needed and not too obstructive, can significantly reduce costs by looking at the lower end of this current range
    - 20KV high voltage supply but jank, still usable but more work to get right and less experimentally accurate
        - allows up to 25mm gap distance, ~1-3mA current capacity, a bit of a stretch for multi-stage
        - Uses AC current and multipliers, followed by smoothing into DC current, definitely less expensive but reduced experimental quality
        - requires more consideration of insulated experimental environment
    - 5-10KV high voltage supply, the budget option
        - Allows for basic experimentation with coronal discharge and ionic wind thrusters, not applicable to full scale models or prototypes
        - Entirely functional experimentally and can maintain coronal discharge but would primarily serve as a stop gap until we could obtain enough funding to put together a more advanced power supply while still being able to do meaningful research
- First thing we need to determine is how we want to handle the early phases of our research and what our “minimum scale” is for a viable protype and what type of currents / gap sizes we want to be working at to properly test our different geometries
    - Joey and eliane this will largely depend on what you want to achieve in preliminary testing with alternate geomotries and the minimum scale of setup we will need to evaluate those experimentally
- This will be the largest obstacle to the progression of our research early on, but I see good and clear ways to move forward with this